

**BUBT**BANGLADESH UNIVERSITY OF
BUSINESS AND TECHNOLOGY

Committed to Academic Excellence

THEORY COURSE OUTLINE

1	Faculty	Faculty of Science and Engineering			
2	Department	Computer Science and Engineering			
3	Program	B.Sc. in CSE			
4	Course Title	Accounting and Management			
5	Course Code	ACT 301			
6	Trimester	Fall-2026			
7	Pre-requisites				
8	Course Type	Core Course/Non-Departmental			
9	Credit Hours	2.30 Hours			
10	Intake/Section	53/1-8 (Day)			
11	Class Schedule	Intake /Section	Class Day	Class Hours	Venue
		53/01	Tuesday	8:15 AM to 9:45AM	B2-319
			Thursday	11:15 AM to 12:45PM	B2-317
		53/02	Tuesday	1:15 PM to 2:45 PM	B2-317
			Thursday	2:45 PM to 4:15 PM	B2-319
		53/03	Sunday	11:15 AM to 12:45PM	B3-905
			Thursday	8:15 AM to 9:45AM	B2-910
12	Campus	Permanent Campus			
13	Course Website				
14	Course Instructor Details	Name: Md. Billal Hossain Designation: Lecturer Specialization: Accounting, Taxation, Cost Accounting, Corporate Governance, IAS-IFRS, Management Accounting. Room No. 501(Building 04) Email: billalbappy57@gmail.com Cell No. 01640888457			
15	Counselling Schedule		Day	Counseling Hours	Venue
		53/01	Tuesday	11:15 AM to 12:15 PM	B4-501
			Thursday	1:15 PM to 2:15 PM	B4-501
		53/02	Tuesday	9:45 AM to 10:45 AM	B4-501
			Thursday	9:45 AM to 10:45 AM	B4-501
		53/03	Sunday	9:45 AM to 10:45 AM	B4-501
			Monday	1:15 PM to 2:15 PM	B4-501
16	Text Book	[1] Accounting Principles, 14 th ed., by Weygandt, J. J., Kimmel, P. D., & Mitchell, J. E., Singapore: John Wiley & Sons. [2] Management, 13 th ed., by Robbins, S. P., & Coulter, M (2017), India: Pearson.			
17	Reference Book	[1] Financial Accounting, 10 th ed., by Libby, R., Libby, P., & Hodge, F. (2019), McGraw Hill Inc. [2] Principles of Management, by Tripathi, P. C. & Reddy, P. N., (2008), Tata McGraw Hill Publishing Company Limited.			

18	Equipment & Aids	Students must carry learning materials (<i>notebook, pen, etc.</i>) in classroom. Borrowing learning materials in exam or classroom from fellow students is strictly prohibited.					
19	Course Rationale	The main objective of this course is to increase application of accounting information in decision making, there developed specific specialization within the subject and to be the future managers. The traditional financial accounting has been used to address the needs of external users as also the statutory requirements of disclosure, while a newly developed branch namely, cost and management accounting focuses on satisfying the data and information requirements of internal users for decision making. In this course we shall find a selection of topics taken from both these branches of accounting and management. Particularly students who do not desire to specialize in the discipline of accounting but still require an overall knowledge of the subject in its totality will benefit much from this course. A very good knowledge and skill in the use of tools and techniques of accounting and management is a prerequisite for becoming a good executive in any organization. This course also provides knowledge to expose the learners to the challenges of managing enterprises – private or public, corporate or small business enterprises, and government or non-government organizations. This course is an introduction to the critical management skills involved in decision-making, planning, organizing, leading, and controlling in an organization.					
20	Course Synopsis	Introduction: Definition of accounting, Three major activities of Accounting, Objectives of accounting, Importance of accounting, Accounting as an information system, accounting & its relationship with CSE; Basic accounting equation, Transaction analysis & its impact on accounting equation; Transaction analysis & its impact on accounting equation; Preparation of Financial Statements; Double entry system: Rules for debit & credit; The recording process: recording transactions in general journal; Recording transactions in general journal; Classifying the transactions, types of accounts in ledger; posting the transactions in ledger, Chart of accounts, preparing trial balance, Timing issues in accounting: Revenue recognition principle & Expense recognition principle; need for adjusting entries, Adjusting entries for Prepayments, Adjusting entries for Accruals, Preparation of Adjusted Trial Balance, Accounting cycle: financial statements, Financial statements analysis & interpretation: ratio analysis.					
21	Course Outcomes (COs)	Upon completing this course student will be able to: CO1: Explain the steps of accounting cycle. CO2: Evaluate the financial performance of an organization using financial report. CO3: Analyze and evaluate financial information relating to product or service costs and contribution to profit					
CO Attainment Process							
CO No.	PO No.	Bloom’s Domain / Level	Knowledge Profile, K	Complex Engineering Problems, WP	Delivery Methods / Activities	Assessment Tools	
CO1	PO2	Cognitive / Understanding	K7	---	• Class Lecture • Audio-Visual	• Attendance & class performance • Class Test • Assignment • Mid Exam • Final Exam	
CO2	PO4	Cognitive / Analyzing	K7	P1			
CO3	PO7	Cognitive / Understanding	K7	P1	• Home Assignment • Class Exercise • Group Discussion		

22	Teaching Strategy	Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. For some cases, multimedia projector will be used for the convenience of the students. Students must participate in classroom discussions for case studies, problems solving and project developments			
23	Lecture Plan				
	Lecture #	Selected Topics	Article (Text book)	Prob./Activities (Text book)	Course Outcome
	1-2 (Week 1)	Accounting in Action An Introduction: Meaning of Accounting and uses of accounting data. Brief History of Accounting. Distinction between bookkeeping and accounting. Accounting and the reader. The Accounting profession. The building blocks of Accounting: Ethics – A fundamental business concept. Generally Accepted Accounting Principles (GAAP). Assumptions. Basic Accounting Equation. Transaction Analysis. Summary of Transactions.	Chapter. 1	Class Test 1 Assignment 1	CO1
	3-4 (Week 02)	Financial Statements: Income Statement – Owner’s Equity Statement – Balance Sheet – Statement of Cash Flows.	Chapter 1 & Hand Notes		CO1
	5-6 (Week 03)	The Recording Process The Recording Process Illustrated: Journalizing and Posting. Preparation of a trial balance, Limitations of a Trial Balance Locating Errors. Timing Issues, Selecting an Accounting Time Period. Fiscal and Calendar Years.	Chapter 2 Hand Notes		CO1
	7-8 (Week 04)	Adjusting the Accounts Accrual vs. Cash Basis Accounting. Recognizing Revenues and Expenses. The Basics of Adjusting Entries: Types of Adjusting Entries. Adjusting Entries for Prepayments. Adjusting Entries for Accruals. Summary of Basic Relationships. The Adjusted Trial Balance and Financial Statements: Preparing the Adjusted Trial Balance. Preparation of Financial Statements.	Chapter 3		CO1
		<i>Class Test</i>			
	9-10 (week 5)	Completing the Accounting Cycle Completion of the Accounting Cycle: Using a Worksheet. Steps in the Preparation of a Work sheet. Preparing Financial Statements from a Work sheet. Preparing Adjusting Entries from a Worksheet. Preparing Closing Entries. Posting Closing Entries. Preparing a Post-Closing Trial Balance. Classified Income Statement. Classified Balance Sheet. Forms of Financial Statements: Multiple Step Income Statement. Single Step Income Statement. Classified Statements. Preparing a Post-Closing Trial Balance. Classified Income Statement. Classified Balance	Chapter 4		CO1

		Sheet. Forms of Financial Statements: Multiple Step Income Statement. Single Step Income Statement. Classified Statements.			
	11-12 (Week 6)	Introduction to Management Definition of Management, Characteristics of Management, Management- Science or Art, The Process of Management (management functions), Decision making, Planning, Organizing, Staffing, Directing, Motivating and Leading, Coordinating, Controlling, Why Study Management? Universality of Management. Explain what a manager is? Distinction between manager and non-manager, Levels of Management, Skill required for managers, Efficiency and Effectiveness of Management, Roles of Management, Why study Management?	Hand Notes		CO2/CO3
	13-14 (Week 07)	Management yesterday and today Early examples of management, Industrial Revolution and its impact on the evolution of management theories / thoughts / approaches. Theories/Approaches of Management – Scientific Management Approach. Administrative Management Approach Henri Fayol's 14 principles, Administrative Management Approach-Ideal Bureaucracy. Systems Approach of Management, Organizational behavioral approach of management, Contingency approach of management.	Hand Notes		
	Midterm Examination– (as per academic calendar)				
	15-16 (Week 08)	Accounting for Merchandising Operations Documents and Terms used in Buying and Selling of Inventories. Types of Inventory Accounting. Differences in Methods of Recording. Record Sales under a Perpetual Inventory System. Sales Returns and Allowances. Sales Discounts. Recording Purchases and Sales under a Periodic Inventory System. Apply the Steps in the Accounting Cycle to a Merchandising Company. Compare a Multiple-Step with a Single-Step Income Statement in a Merchandising Company.	Chapter 5		CO2/CO3
	17-18 (Week 09)	Inventories Inventories Classification and Determination of Inventory. Inventory Cost Flow methods and their Financial Effects. Inventory Cost Flow Methods to Perpetual Inventory Records. FIFO, LIFO and Average Cost. Methods of Estimating Inventories. Effects of Inventory Errors on the Financial Statements. Statement Presentation and Analysis of	Hand notes & Chapter 6		CO2/CO3

		Inventory. Preparation of Comparative Income Statement for different Cost Flow Methods Used in Accounting.			
19-20 (Week 10)		Accounting Information Systems Accounting Information Systems: Basic Concepts of AIS, Computerized Accounting Systems, Manual Accounting System, and Subsidiary Ledger. Mechanistic and Organic Organizations, Formalization. Sales Journal, Cash Receipts Journal, Purchases Journal, Cash Payments journal, Effects of Special Journals on the General Journal. Fraud, Internal Control and Cash: Fraud, Internal Control and Cash: Principles of Internal Control Activities, Limitations of Internal Control, Cash Receipts, Disbursements and Petty Cash. Bank Reconciliation statement.	Hand notes & Chapter 7 & 8	Assignment 2	CO2/CO3
21-22 (Week 11)		Accounting for Receivables Types of Receivables, Recognizing Accounts Receivables, Valuing Accounts Receivables, Disposing of Accounts Receivables, Computing Interest, Notes Receivables, Valuation and Disposition of Notes Receivables, Statement Presentation and Analysis.	Hand notes Chapter 9		CO2
23-24 (Week 12)		Financial Statement Analysis Financial Statement Analysis. Basics of Financial Statement Analysis. Need for Comparative Analysis. Tools of Analysis. Horizontal Analysis. Balance Sheet, Income Statement and Retained Earnings Statement Vertical Analysis. Balance Sheet and income Statement. Ratio Analysis. Liquidity Ratios, Profitability Ratios. Solvency Ratios. Summary of Ratios. Earning Power and Irregular Items. Discontinued Operations. Extraordinary Items. Changes in Accounting Principle. Comprehensive Income. Quality of Earnings. Alternative Accounting Methods. Pro Forma Income. Improper Recognition.	Chapter 18	Assignment 3	CO3
25-26 (Week 13)		Management Functions Decision Making – Decision making process, steps of decision-making process. Three perspectives of decision Making-Rational Decision-making, Bounded Rational Decision-making, Role of Intuition in Decision Making. Types of problems and types of decisions, Decision-making conditions. Foundations of Planning-Definition of Planning, Formal Planning, Purposes/Importance of Planning, Plans, Types of Plans, Goals, Types of Goals, Approaches to Establishing Goals, Steps in Goal	Hand notes		CO3

		Setting, Strategy and Strategic Management. MBO-definition, how does MBA work? Definition of Organizing, Organization, Organizational Structure and Design, Key elements of organization- Work Specialization, Chain of Command, Span of Control. Departmentalization, Centralization and Decentralization of Authority, Mechanistic and Organic Organizations, Formalization. Staffing – Definitions, Functions of Staffing, Importance of Staffing. Sources of Recruitment, Steps of the Selection Process.			
	27-28 (Week 14)	Directing -Definition, Principles of Effective Direction, Conditions of an Order to be acceptable. Motivation – Definition, Theories of Motivation– Maslow’s Need Hierarchy Theory of Motivation, McGregor’s Theory X and Theory Y of Motivation, Herzberg’s Two-Factor Theory of Motivation, McClell and’s Need for Achievement Theory of Motivation. Leading – Definition, Managers vs. Leaders, Leadership Traits. Leadership Approaches-Behavioral Approach of Leadership. Coordinating-Definition, Unity of action among the employees, Groups, and Departments. Controlling-Definition, steps in Controlling Process, Need for Control, Types of Controlling Methods. Social Responsibility and Management-Social responsibility, Classical view and socioeconomic view, Arguments for and Against Social Responsibility, Social responsibility and economic performance, the greening of management, Values-based management.	Hand notes	Class Test 3	CO1, CO2, CO3
		Final Examination- (as per academic calendar)			
24	Assessment and Marks Distribution:	Students will be assessed on the basis of their overall performance in all the exams, quizzes, and class participation. Final numeric reward will be the compilation of (tentative): ❖ Class Tests (15%) ❖ Assignment and Class Presentation (10%) ❖ Attendance (5%) ❖ Mid-Term Test (30%) ❖ Final Exam (40%)			
25	CO Assessment Criteria	Assessment methods of COs are given below:			
		CLOs	Statements		
		CLO1	To define bookkeeping and accounting and evaluate the global context for taking managerial actions of planning, organizing, leading and controlling.		
		CLO2	To explain the general purposes and functions of accounting and integrate management principles in to management practices.		
		CLO3	To explain the differences between management and financial accounting and assess managerial practices and choices relative to ethical principles and standards.		
		CLO4	To describe the main elements of financial accounting information – assets, liabilities, revenue and expenses and specify how the managerial tasks of planning, organizing, and controlling can be executed in a variety of circumstances.		

	Program Outcomes (POs) are narrower statements that describe what students are expected to know and be able to do by the Time of graduation. These relate to the knowledge skills and attitudes that students acquire while progressing through the program. The students of the B.Sc. in CSE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation. PO1–Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems. PO2–Problem analysis: Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences. PO3–Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns. PO4–Investigation: Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions. PO5–Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations. PO6–The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice. PO7–Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development. PO8–Ethics: Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice. PO9–Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings. PO10–Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions. PO11–Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work as a member or a leader of a team to manage projects in multidisciplinary environments. PO12–Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.
31	Knowledge Profile (WK) WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline. WK2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline. WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline. WK5: Knowledge that supports engineering design in a practice area.

	WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline. WK7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability. WK8: Engagement with selected knowledge in the research literature of the discipline.		
32	Range of Complex Engineering Problem Solving		
	Complex Engineering Problems have characteristic WP1 and some or all of WP2 to WP7: WP1–Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach WP2–Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering and other issues WP3–Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models WP4–Familiarity of issues: Involve infrequently encountered issues WP5–Extent of applicable codes: Are outside problems encompassed by standards and codes of practice for professional engineering WP6–Extent of stakeholder involvement and conflicting requirements: Involve diverse groups of stakeholders with widely varying needs WP7–Interdependence: Are high level problems including many component parts or sub-problems		
33	Range of Complex Engineering Activities		
	Attribute Complex activities means (engineering) activities or projects that have some or all of the following characteristics: EA1–Range of resources: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies) EA2–Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues EA3–Innovation: Involve creative use of engineering principles and research based knowledge in novel ways EA4–Consequences for society and the environment: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation EA5–Familiarity: Can extend beyond previous experiences by applying principles-based approaches		
34	Social & Moral Capital		
	Our promises are based on the three cardinal principles: (a) What we do believe (b) What we do practice, and (c) What we will promote However, students are advised to undertake the following commitments for moral development.		
	1. To be punctual and attentive in class 2. To maintain inclusive learning environment 3. To ensure mutual respect 4. To be cooperative in group learning. 5. To be innovative and Creative 6. To follow dress code and wearing ID card	8. Try to follow and review day to day class 9. To avoid conspiracy 10. To prioritize honesty & faith 11. To be motivated for asking question and encourage feedback 12. To develop attitude for speaking in English	15. To be sincere for class preparation 16. Do not forget to switch-off the cellphone in class 17. Do not forget to carry course pack and learning stuffs in class 18. To maintain loyalty and trust to the university

	7. To be always proactive	13. Do not ignore to carry out any assignments or commitments 14. To be clean and decent in all levels.	19. Must avoid unfair means and plagiarism in exam, reports and assignments 20. Must maintain eco-friendly environment in the campus.
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